

Late Breaking Results: Coulomb Force-Based Routability-Driven Placement Considering Global and Local Congestion*

Jihai Meng¹, Shaohong Weng², Zhijie Cai³, Yilu Chen¹, Zhifeng Lin^{1,3}, and Jianli Chen³

¹Center for Discrete Mathematics and Theoretical Computer Science, Fuzhou University, Fuzhou 350108, China

²College of Computer and Data Science, Fuzhou University, Fuzhou 350108, China

³State Key Laboratory of Integrated Chips and Systems, Fudan University, Shanghai 200433, China
220320049@fzu.edu.cn;linzhifeng@fzu.edu.cn;chenjianli@fudan.edu.cn

ABSTRACT

Placement is a critical stage for VLSI routability optimization. A placement engine without considering the layout congestion might lead to poor solutions with routing failures. This paper introduces a Coulomb force-based global placement framework that addresses global and local routing congestions. We first present a routing path-based cell padding strategy for local congestion mitigation. Then, we construct a routability-aware placement model that utilizes virtual Coulomb forces to eliminate crucial global congestion. Compared with a leading academic placer, RePlAce, and the advanced commercial tool, Innovus, the experimental results on industrial benchmark suites show that our proposed algorithm achieves the best routability within the shortest runtime.

1 INTRODUCTION

Placement is an essential stage in VLSI design flow. It determines the physical locations of all movable blocks and thus has significant impacts on the overall power consumption, area utilization, and timing closure. As the feature sizes keep scaling, the design complexity of modern circuits has increased dramatically, incurring significant challenges to the layout routability. Thus, developing an effective and efficient placement algorithm for routability optimization is desirable.

Typically, existing routability-driven placement algorithms can be categorized into three types: cell inflation [1], white space allocation [2], and analytical algorithms [3]. Cell inflation and white space allocation methods optimize congestion by spreading cells in congested local areas, and they often cannot address global congestion. Analytical algorithms aim to integrate congestion information into the placement objective, utilizing cell diffusion to optimize routing congestion. However, this approach lacks specific handling of localized congestion areas.

In this paper, we propose a Coulomb force-based placement framework to eliminate both global and local congestion. Compared with a leading academic placer, RePlAce, and the state-of-the-art commercial tool, experimental results on industrial benchmarks show that our algorithm achieves the best routability, with an average of 1.25× and 2.32× speedups.

2 PROPOSED METHOD

2.1 Local Congestion Mitigation

Local congestion typically arises from the significant clustering of cells in the GP stage. To mitigate such congestion, a common approach involves adjusting the sizes of cells during the GP stage to amplify the gradient thrust of the density function, thereby dispersing clustered cells and alleviating local congestion. Previous modeling approaches often focused solely on internal factors of GCells (Routing Grids), such as pin density and congestion magni-

tude. However, external factors related to GCells, such as congestion areas traversed by interconnections, may also constitute hidden contributors to congestion risk. Therefore, to more accurately perceive cell padding due to congestion, we decompose the padding ratio (Pr) linearly into external ($Gout$) and internal (Gin) factors of GCells:

$$Pr = Lcoef \cdot Gin + Ncoef \cdot Gout + intercept, \quad (1)$$

where $Lcoef$ and $Ncoef$ signifies the weight, $intercept$ represents adjustable parameters. Pin densities and congestion information are considered as Gcell internal factors:

$$Gin = \log_2(\max(PinDen \cdot Pcoef + Gcg \cdot Gcoef, 1)), \quad (2)$$

where $PinDen$ stands for pin densities, Gcg represents gcell congestion information, $Pcoef$ and $Ocoef$ represent the weight coefficient. The congestion of the gcell traversed by cell-connected two-pin nets will be considered as an external factor to the gcell:

$$Gout = \frac{1}{Netnum} \sum_{net_j \in cell} netpadding_j, \quad (3)$$

$$netpadding_j = \max_{g \in path_j} \frac{overflow_g}{capacity_g}, \quad (4)$$

where $Netnum$ reflects the number of two-pin nets connected to a cell, $netpadding_j$ represents the maximum gcell congestion after normalization within a net path j .

2.2 Global Congestion Elimination

Based on optimization theory, we introduce an implicit optimization function with a global routing congestion (GRC) objective:

$$\min_{(x,y)} \ell(x,y) = W(x,y) + \lambda D(x,y) + \beta GRC(x,y), \quad (5)$$

where $GRC(x,y)$ denotes routing congestion and β is the coefficient. The directional superposition of the objective function with global congestion descend (GCD) is as follows:

$$TarDiv = NetGrad + \lambda DenseGrad + \beta GCD, \quad (6)$$

where $NetGrad$ and $DenseGrad$ stand for the wirelength gradient and electric field force gradient in the ePlace [4] algorithm. GCD is termed the congestion descent direction.

To calculate GCD, we envision another virtual force within the electrostatic field guiding cells toward reduced congestion:

$$Cforce_i = direction_i \cdot \mu \cdot \frac{CellArea_i \cdot ImArea_i}{d^2}, \quad (7)$$

where $Cforce_i$ as the Coulombic force exerted on cells i by its corresponding virtual cells, $direction_i$ signifies the force direction. The $direction$ will be set to positive 1 when pre-movement computationally up or to the right will reduce congestion, and negative 1 if not. $CellArea_i$ represents the cells's area, μ indicates the electrostatic coefficient and $ImArea_i$ denotes the area associated with virtual cells i , d is the distance between cell i and virtual cell i . The determination of $ImArea_i$ will be achieved through a nonlinear transformation:

$$ImArea_i = \varepsilon \cdot Npc_i, \quad (8)$$

$$Npc_i = \sqrt{\max(NetsPos_i^{ori} - \xi \cdot NetsNeg_i^{ori}, 0)}, \quad (9)$$

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Table 1: Comparison of HOF, VOF, WL, and RT Among RePlAce, Innovus, and Ours on Industrial Benchmarks

Benchmark	Statistic			RePlAce				Innovus				Ours			
	#Stdts	#Nets	#Macros	HOF(%)	VOF(%)	WL	RT(s)	HOF(%)	VOF(%)	WL	RT(s)	HOF(%)	VOF(%)	WL	RT(s)
Cir.A	122k	193k	22	0.92	1.33	3238951	449	0.88	0.21	3724999	1006	0.36	0.02	3078260	217
Cir.B	974k	980k	46	0.9	7.8	28031430	4296	0.4	0.96	27003190	3887	0.04	0.34	28630480	3231
Cir.C	1731k	1746k	115	0.35	0.18	50469630	6109	0.21	0.17	43383470	8703	0.2	0.13	44373710	5205
Cir.D	1228k	1296k	70	4.44	14.84	33373000	3350	0.67	5.83	30304690	8005	0.8	0.85	31001120	3643
Cir.E	1232k	1300k	7	3.34	14.4	33464500	3388	0.15	0.39	30524130	7643	0.3	0.7	32703430	3426
Cir.F	2146k	2155k	152	0.22	0.15	27843220	6760	0.03	0.04	27207150	8257	0.13	0.12	27145420	5360
Cir.G	3669k	3716k	196	0.53	0.3	79324310	9730	0.28	0.18	75166950	18322	0.22	0.12	76343520	7619
Cir.H	4487k	4510k	113	2.68	5.25	63992550	12330	1.37	1.45	63040820	20848	0.83	0.83	64138130	11017
Cir.I	4756k	4768k	161	4.56	2.12	57961450	9568	2.48	1.13	49216310	26315	2.38	0.43	49382140	8154
Cir.J	4377k	4401k	244	1.63	0.32	66321040	10890	0.16	0.06	65027090	20569	0.39	0.3	65196310	9170
Average	2472k	2506k	112	1.957	4.675	1.060	1.250	0.663	1.042	1.001	2.324	0.565	0.387	1.000	1.000

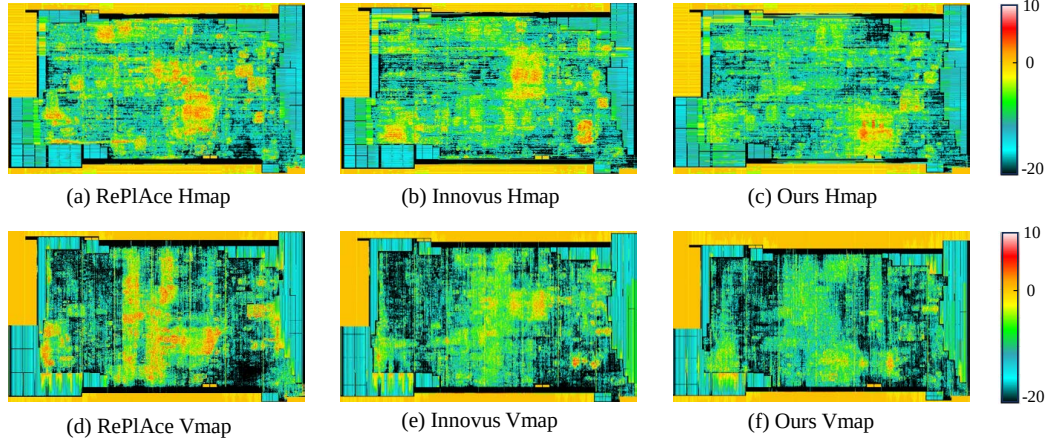


Figure 1: Congestion maps for Cir. H reported by Innovus global router. (a), (b) and (c) are the horizontal congestion maps of the placement layouts generated by RePlAce, Innovus, and our algorithm. (d), (e) and (f) represent the vertical congestion maps corresponding to the placement results produced by RePlAce, Innovus, and our approach, respectively.

where $ImArea_i$ signifies the transformation of the ultimate effect into the area of the corresponding virtual cell, ε represents the area convert coefficient, Npc_i denotes the final force of cell i after pre-movement, $NetsPos_i^{ori}$ denotes the positive force in the different orientation of the nets connected by cell i , and $NetsNeg_i^{ori}$ represents the corresponding escape resistance, ξ represents the weight used to consider resistance. Positive force is generated by the congestion calculation of the two-pin net inside the Gcell when the cell pre-movement and resistance force is generated by the other two-pin nets in the rest of the cell. Commonly, when a two-pin net of a cell passes vertically through a Gcell, the NetsPos, and NetsNeg for the left and right path directions are computed to evaluate net movement in the X-axis.

After modeling the positions and areas of virtual cells and obtaining their coordinates, we derive Coulomb forces from these virtual cells. The calculation in the X-axis is as follows:

$$GCD_X = \kappa \cdot forceH, \quad (10)$$

where κ represents the gradient conversion coefficient, $forceH$ represents Coulomb force on H direction. Overall, the cell's congestion descent direction is formed by concatenating the X and Y-axis:

$$GCD = (GCD_X, GCD_Y) \quad (11)$$

3 EXPERIMENTAL RESULTS

We implemented our placer in the C++ programming language and conducted experiments on industrial benchmarks. Table 1 gives the benchmark statistics, where “#Stdts”, “#Nets”, and “#Macros” denote the number of movable standard cells, nets, and preplaced macros, respectively. To evaluate the effectiveness of the proposed placer, we compared our algorithm (named “Ours”) with the state-of-the-

art academic placer, RePlAce [1], and the leading commercial tool, Innovus. We ran all the experiments on the same Linux workstation, Intel Xeon with 2.4GHz using eight threads, and utilized the Innovus global router to report the routability of the placement layout.

From Table 1, it can be seen that our algorithm achieves the best horizontal and vertical routing overflows without sacrificing the routed wirelength. Our routability improvement mainly comes from considering both global and local congestion. Our placer achieves respective 1.2× and 2.3× speedups for the runtime compared with RePlAce and Innovus. Figure 1 gives the horizontal and vertical congestion maps of the placement layouts produced by RePlAce, Innovus, and our algorithm. Overall, the experimental results show that our proposed framework is effective and efficient for the addressed routability-driven global placement problem.

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